Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Understanding the Derivative Vincent Knight 1984/02/14

1 Introduction

Differentiation is a concept of Mathematics studied in Calculus. There is an ongoing discussion as to who was the first to define differentiation: Leibniz or Newton [**?**].

Differentiation allows for the calculation of the slope of the tangent of a curve at any given point as shown in Figure 1.

[domain=-2:2, color=blue] plot (, 1 - ()^2)$node[above = .5cm, right, color = blue]$f(x)=1-x^2; [domain=-2:2, color=red] plot(,-1 * + 1.25) node[above = .5cm, right, color=red] Tangent at $x = .5$; [thick, ->] (-3,0) – (3,0) node [above] x; [thick, ->] (0,-3) – (0,3) node [right] y; at (.5,.75) •;

Gambar 1: The plot of $f(x) = 1 - x^2$ with a tangent at $x = .5$.

Differentiation is now a technique taught to mathematics students throughout the world. In this document I will discuss some aspects of differentiation.

2 Exploring the derivative using Sage

The definition of the limit of $f(x)$ at $x = a$ denoted as $f'(a)$ is:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{1}$$

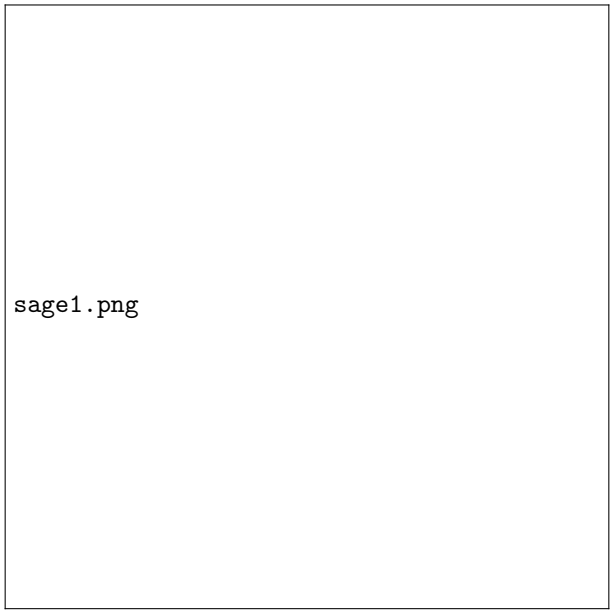
The following code can be used in sage to give the above limit:

```
def illustrate(f, a):
    """
    Function to take a function and illustrate the limiting definition of a derivative at a given po
    """
    lst = []
    for h in srange(.01, 3, .01):
     lst.append([h,(f(a+h)-f(a))/h])
    return list_plot(lst, axes_labels=['$x$','$\\frac{f(%.02f+h)-f(%.02f)}{h}$' % (a,a)])
```

If we want to plot the tangent at a point α to a function we can use the following:
y=ax+b(definition of a straight line)
f'(a)x+b(definition of the derivative)
f'(a)x+f(a)-f'(a)a(we know that the line intersects f at $(a, f(a))$

We can combine this with the approach of the previous piece of code to see how the tangential line converges as the limiting definition of the derivative converges:

```
def convergetangentialline(f, a, x1, x2, nbrofplots=50, epsilon=.1):
    """
    Function to make a tangential line converge
    """
    clrs = rainbow(nbrofplots)
    k = 0
```



Gambar 2: The derivative of $f(x) = 1 - x^2$ at $x = .5$ converging to -1 as $h \to 0$.

```
h = epsilon
p = plot(f, x, x1, x2)
while k < nbrofplots:
    tangent(x) = fdash(f, a, h) * x + f(a) - fdash(f, a, h) * a
    p += plot(tangent(x), x, x1, x2, color=clrs[k])
    h += epsilon
    k += 1
return p
```

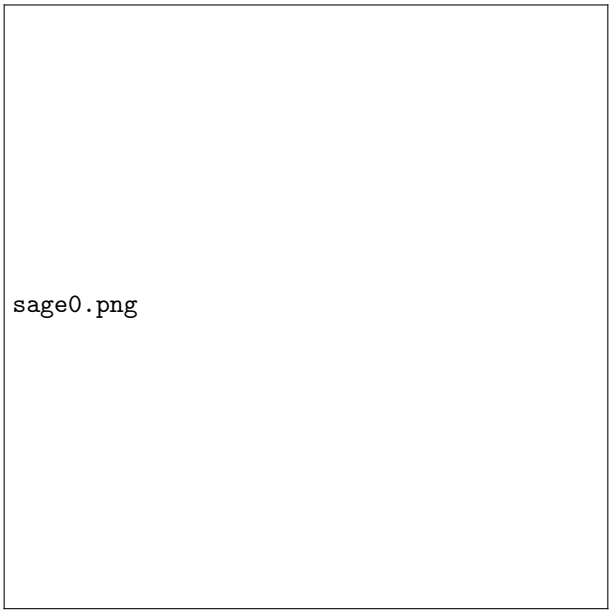
The plot shown in Figure 3 shows how the lines shown converge to the actual tangent to $1 - x^2$ as $x = 2$ (the red line is the 'closest' curve).

Note here that the last plot is given using the **real** definition of the derivative and not the approximation.

3 Conclusions

In this report I have explored the limiting definition of the limit showing how as $h \to 0$ we can visualise the derivative of a function. The code involved `https://sage.maths.cf.ac.uk/home/pub/18/` uses the differentiation capabilities of Sage but also the plotting abilities.

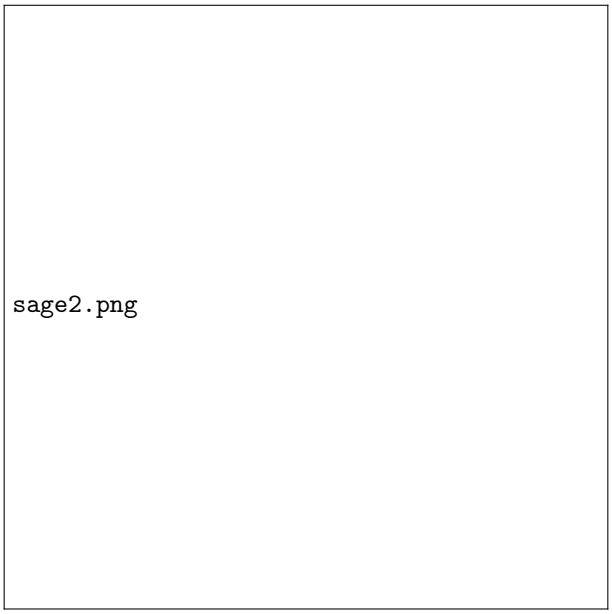
There are various other aspects that could be explored such as symbolic differentiation rules. For example:



Gambar 3: Lines converging to the tangent curve as $h \to 0$.

$$\frac{dx^n}{dx} = (n+1)x^n if x \neq -1$$

Furthermore it is interesting to not that there exists some functions that **are not** differentiable at a point such as the function $f(x) = \sin(1/x)$ which is not differentiable at $x = 0$. A plot of this function is shown in Figure 4.



Gambar 4: None differentiable function at $x = 0$.